

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO: 2520030071

NAME: G. Hemanth Venkatesh

Query 1: - CREATE DATABASE

```
1 2520030071-G. Hemanth Venkatesh
2 CREATE DATABASE IF NOT EXISTS joins_db;
3 USE joins_db;
```

Output: -

	#	Time	Action	Message	Duration / Fetch
✓	1	13:55:57	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected	0.031 sec
✓	2	13:56:01	USE joins_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLES

```
1 CREATE TABLE class (
2     id INT,
3     name VARCHAR(30)
4 );
5 CREATE TABLE class_info (
6     id INT,
7     address VARCHAR(30)
8 );
```

Output: -

	#	Time	Action	Message	Duration / Fetch
✓	3	13:58:04	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.125 sec
✓	4	13:58:07	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected	0.047 sec

Query 3: -INSERT VALUES

```

1 25200:30478-G. Hemanth Venkatesh
2 INSERT INTO class VALUES
3 (1, 'G. Hemanth Venkatesh'),
4 (2, 'Rohit Verma'),
5 (4, 'Vivek Sharma');
6
7 INSERT INTO class_info VALUES
8 (1, 'HYDERABAD'),
9 (2, 'DELHI'),
10 (3, 'CHENNAI');

```

Output: -

	#	Time	Action	Message	Duration / Fetch
✓	7	14:01:03	INSERT INTO class VALUES (1, 'G. Hemanth Venkatesh'), (2, 'Rohit Verma'), (4, 'Vivek Sharma')	3 row(s) affected	0.031 sec
✓	8	14:01:08	INSERT INTO class_info VALUES (1, 'HYDERABAD'), (2, 'DELHI'), (3, 'CHENNAI')	3 row(s) affected	0.062 sec

Query 4: - CROSS JOIN

```

1 25200390478-G. Hemanth Venkatesh
2 SELECT *
3 FROM class
4 CROSS JOIN class_info;

```

Output: -

✓	17	14:04:24	SELECT * FROM class CROSS JOIN class_info	9 row(s) returned	0.000 sec / 0.000 sec
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Result Grid   Filter Rows: Export: 				
	id	name	id	address
▶	1	G. Hemanth Venkatesh	1	HYDERABAD
	2	Rohit Verma	1	HYDERABAD
	4	Vivek Sharma	1	HYDERABAD
	1	G. Hemanth Venkatesh	2	DELHI
	2	Rohit Verma	2	DELHI
	4	Vivek Sharma	2	DELHI
	1	G. Hemanth Venkatesh	3	CHENNAI
	2	Rohit Verma	3	CHENNAI
	4	Vivek Sharma	3	CHENNAI

Result 1 × 

Query 5: - INNER JOIN

```

1 25200030478-G. Hemanth Venkatesh
2 SELECT c.id, c.name, ci.address
3 FROM class c
4 INNER JOIN class_info ci ON c.id = ci.id;

```

Output: -

✓	18	14:08:29	SELECT * FROM class INNER JOIN class_info O...	2 row(s) returned	0.000 sec / 0.000 sec
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Result Grid   Filter Rows:				
	id	name	id	address
▶	1	G. Hemanth Venkatesh	1	HYDERABAD
	2	Aditya Rao	2	DELHI

Query 6: -INNER JOIN

Output: -

✓	19	14:10:36	SELECT class.name, class_info.address FRO...	2 row(s) returned	0.000 sec / 0.000 sec
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Result Grid   Filter Rows:		
	name	address
▶	G. Hemanth Venkatesh	HYDERABAD
	Aditya Rao	DELHI

Query 7: - NATURAL JOIN

Output: -

✓	20	14:12:02	SELECT * FROM class NATURAL JOIN class_info	2 row(s) returned	0.000 sec / 0.000 sec
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Result Grid   Filter Rows:			
	id	name	address
▶	1	G. Hemanth Venkatesh	HYDERABAD
	2	Aditya Rao	DELHI

Query 8: - LEFT OUTER JOIN

Output: -

✓	21	14:14:30	INSERT INTO class VALUES (5,'Nihitha')	1 row(s) affected	0.031 sec
✓	22	14:14:36	INSERT INTO class_info VALUES (7,'BENGALURU'),(8,'PUNE')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.046 sec

Result Grid					Filter Rows:	Export:
	id	name	id	address		
▶	1	G. Hemanth Venkatesh	1	HYDERABAD		
	2	Aditya Rao	2	DELHI		
	3	Neha Sharma	NULL	NULL		
	5	Nihitha	NULL	NULL		

Query 9: - LEFT JOIN

Output: -

✓	23	14:16:23	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id	4 row(s) returned	0.000 sec / 0.000 sec
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Result Grid					Filter Rows:	Export:
	id	name	id	address		
▶	1	Adama Samarth	1	HYDERABAD		
	2	Aditya Rao	2	DELHI		
	3	Neha Sharma	NULL	NULL		
	5	Nihitha	NULL	NULL		

Query 10: - LEFT JOIN

Output: -

✓	24	14:18:42	SELECT * FROM class_info LEFT JOIN class ON class_info.id=class.id	4 row(s) returned	0.000 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

Result Grid					Filter Rows:	Export:
	id	address	id	name		
▶	1	HYDERABAD	1	G. Hemanth Venkatesh		
	2	DELHI	2	Aditya Rao		
	7	BENGALURU	NULL	NULL		
	8	PUNE	NULL	NULL		

Query-11

Result Grid				
	id	name	id	address
▶	1	G. Hemanth Venkatesh	1	HYDERABAD
	2	Aditya Rao	2	DELHI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Query-12: Union

25290930478-G. Hemanth Venkatesh

```
CREATE TABLE first_table(  
    roll_no INT,  
    name VARCHAR(30)  
);  
  
CREATE TABLE second_table(  
    roll_no INT,  
    name VARCHAR(30)  
);  
  
INSERT INTO first_table VALUES  
    (101, 'Rahul'),  
    (102, 'Vikram');  
  
INSERT INTO second_table VALUES  
    (102, 'Vikram'),  
    (103, 'Arjun');
```

✓	28	14:25:46	CREATE TABLE first_table(roll_no INT, name VARC...	0 row(s) affected	0.047 sec
✓	29	14:25:50	CREATE TABLE second_table(roll_no INT, name VA...	0 row(s) affected	0.031 sec
✓	30	14:25:54	INSERT INTO first_table VALUES (101,'Rahul'), (10...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.046 sec
✓	31	14:25:58	INSERT INTO second_table VALUES (102,'Vikram'),...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.047 sec

```
SELECT * FROM first_table  
UNION  
SELECT * FROM second_table;
```

✓	32	14:27:30	SELECT * FROM first_table UNION SELECT * FR...	3 row(s) returned	0.031 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

Result Grid		 Filter Rows:
	roll_no	name
▶	101	Rahul
	102	Vikram
	103	Arjun

Query 13: - UNION ALL

2529030478-G. Hemanth Venkatesh

```
SELECT * FROM first_table  
UNION ALL  
SELECT * FROM second_table;
```

✓	33	14:28:55	SELECT * FROM first_table UNION ALL SELECT ...	4 row(s) returned	0.031 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

Result Grid		 Filter Rows:
	roll_no	name
▶	101	Rahul
	102	Vikram
	102	Vikram
	103	Arjun

Query 14: -INTERSECT (MYSQL DO NOT SUPPORT INTERSECT DIRECTLY)

```
SELECT first_table.*
FROM first_table
INNER JOIN second_table
ON first_table.id = second_table.id;
```

Output: -

✓ 34 14:32:12 SELECT first_table.* FROM first_table INNER JOI... 1 row(s) returned

	id	name
▶	2	Manasvi

Query 15: -MINUS (MYSQL DO NOT SUPPORT MINUS DIRECTLY)

```
SELECT first_table.*
FROM first_table
LEFT JOIN second_table
ON first_table.id = second_table.id
WHERE second_table.id IS NULL;
```

Output: -

✓ 35 14:34:55 SELECT first_table.* FROM first_table LEFT JOIN... 1 row(s) returned

ROLL NO: 2520030071-G. Hemanth Venkatesh

Query 16: -Generate a report showing all students and address availability status.

-- 2520030071-G. Hemanth Venkatesh

```
SELECT c.id,
       c.name,
       CASE
         WHEN ci.address IS NULL
         THEN 'Address Missing'
         ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci
ON c.id=ci.id;
```

Output: -

✓ 37 14:37:51 SELECT c.id, c.name, CASE WH... 3 row(s) returned 0.000 s

	id	name	Status
▶	1	Akshay	Address Available
	2	Manasvi	Address Available
	3	Ramya	Address Missing

Query 17: -FULL OUTER JOIN (MYSQL DO NOT DIRECTLY SUPPORT FULL OUTER JOIN)

ROLL NO: 2520030071-G. Hemanth Venkatesh

```
-- 2520030071-G. Hemanth Venkatesh
```

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL

UNION

SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

Output: -

✓ 38 14:41:25 SELECT * FROM class LEFT JOIN class_info ON... 5 row(s) returned 0.000 sec / 0.000 sec

	id	name	id	address
▶	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE